

FROM REPRESENTATION TO ACT: FOUNDATIONS FOR A REVISED EPISTEMOLOGICAL PARADIGM

The Conceptual Act, Scientific Models, and the Problem of Objectivity

Abstract

The representational paradigm has played a fundamental role in modern epistemology and cognitive science. According to this paradigm, cognition is understood primarily as the manipulation of representations that preserve, reproduce, or model aspects of an external or internal reality. Scientific theories are consequently interpreted through models, concepts are treated as structured representations, and the study of cognition frequently seeks to reconstruct mental processes through verbal reports or other explicit descriptions.

The preceding paper proposed an alternative conception of the concept as an act of structuring experience. That act was provisionally analyzed through three interconnected moments: selection, identification, and projection. The present paper develops the epistemological consequences of this proposal.

Three problems are examined. First, the idealization inherent in scientific models is reconsidered. If models are attempts to represent reality through static structures, their inevitable simplifications need not constitute an epistemological defect; they may reflect the difference between a dynamic act and the static form in which that act is represented. Second, the methodological problem of verbalizing thought is reconsidered. If primary understanding is an embodied and situated act, verbal reports should not be treated as transparent access to cognition itself. They provide evidence concerning the subject's reconstruction and reflection upon an act. Third, the problem of objectivity is addressed. Rather than defining objectivity exclusively as correspondence between representation and an independent reality, this paper proposes to investigate objectivity through the invariance of successful conceptual acts under the expansion of practice.

This approach does not reject scientific models, formal descriptions, representations, or experimental methods. It proposes a change in their epistemological status. They are understood as instruments for stabilizing, communicating, testing, and extending acts of understanding. The resulting framework seeks to establish a relation between knowledge and action without reducing knowledge to subjective utility or abandoning the possibility of objective comparison.

Keywords: epistemology, concept, act, representation, scientific models, idealization, understanding, objectivity, cognition, action.

1. Introduction: From Representation to Act

Modern epistemology and cognitive science have frequently approached cognition through the notion of representation.

To know something is commonly understood as possessing a representation of it. To reason is to manipulate representations. To construct a scientific theory is to formulate representations or models that capture relevant aspects of reality. To understand a concept is to possess or activate an appropriate conceptual structure.

This framework has generated enormous intellectual and scientific achievements.

Nevertheless, the representational paradigm encounters a recurring difficulty: the representation itself does not explain the act through which it becomes meaningful and operative.

A map does not explain the act of navigation. A scientific model does not by itself explain the act of using that model to investigate a phenomenon. A verbal description of a solution does not necessarily reproduce the act through which the solution was discovered. A neural representation does not automatically explain the meaning of the object toward which cognition is directed.

The preceding paper proposed that this difficulty may arise because the concept has been treated primarily as an object rather than as an act.

The present paper develops that proposal.

The central claim is not that representations are unreal or useless. Rather, representations are considered to be **stabilized forms within broader processes of conceptual activity**.

The fundamental epistemological question consequently shifts. Instead of asking only: *"Does a representation correspond to reality?"* we may also ask: *"What does the representation enable a subject to do, recognize, anticipate, test, and correct?"*

This question does not eliminate the problem of truth. It relocates part of the analysis from static correspondence toward situated activity.

2. The Conceptual Act as the Basis of the Analysis

The first paper proposed understanding the concept as an act of structuring experience.

This act contains three analytically distinguishable moments:

1. **Selection** — isolating something as relevant within the continuous field of experience;
2. **Identification** — relating what has been selected to a category, previous experience, sign, or meaningful order;
3. **Projection** — anticipating possible interactions, consequences, or actions.

These moments should not be interpreted as rigid chronological stages. In concrete cognition, selection may already be influenced by anticipation, while identification may alter what becomes salient. Projection can therefore affect subsequent selection. The threefold structure is consequently better understood as a **dynamic organization of conceptual activity**.

The epistemological importance of this proposal is that it places understanding within a process. The concept is no longer something that exists independently of its use and is subsequently applied to experience. Instead, conceptuality emerges in the very organization of experience.

This allows us to reconsider several classical epistemological problems from a common starting point. Scientific models become stabilized instruments of conceptual orientation. Verbal reports become reconstructions of acts. Objectivity becomes a property of acts that remain successful beyond their initial conditions of formation.

3. Scientific Models and the Problem of Idealization

3.1. The Inevitability of Idealization

Scientific models are rarely literal copies of the phenomena they describe. Scientific practice routinely employs idealizations: ideal gases, point masses, frictionless surfaces, infinitely large populations, massless strings, simplified geometries, abstract computational agents. Such entities do not necessarily exist in the form represented by the model.

This raises a classical epistemological question: if a scientific model does not correspond literally to reality, in what sense can it provide knowledge of reality?

One possible response is to treat scientific models as useful fictions. Another is to regard idealization as an unavoidable imperfection of scientific representation.

The act-oriented approach suggests a different interpretation. Idealization may arise because the process being modeled is dynamic, contextual, and practically situated, whereas the model must stabilize selected relations within a relatively fixed structure. The model therefore necessarily abstracts. Its abstraction is not merely a failure to reproduce reality. It is also what makes the model usable.

3.2. Model as Instrument of Orientation

A model can be understood not primarily as a picture of reality but as a structured instrument for performing an epistemic act. Its function may include: identifying relevant variables, isolating relations, anticipating outcomes, organizing observations, guiding experiments, and determining possible interventions.

The value of a model therefore cannot be measured solely by asking whether every element corresponds literally to an element of reality. A more appropriate question is: *"What forms of successful understanding and action does the model make possible?"*

This does not imply that any useful model is thereby true. A model may be useful within one domain and fail outside it. It may generate successful predictions under restricted conditions while becoming misleading when those conditions change. Consequently, usefulness must be combined with the possibility of extension, testing, correction, and comparison. The model's epistemic value lies in its ability to participate in a broader practice of inquiry.

4. The Static Model and the Dynamic Act

The distinction between model and act is fundamental.

A model is relatively stable. An act unfolds. A model can be stored, communicated, reproduced, and examined independently of the particular moment in which it was produced. An act is situated in time and depends upon a concrete relation between a subject, a situation, and possible consequences.

This distinction explains why models inevitably simplify. They transform a dynamic process into a form that can be manipulated as an object. The simplification is simultaneously their limitation and their strength.

A map omits enormous quantities of information precisely so that orientation becomes possible. A structural calculation idealizes material behavior so that a complex physical system can be analyzed. A biological model isolates selected variables so that relations that would otherwise remain obscure can become tractable.

Thus: **the incompleteness of a model does not necessarily invalidate the model; it may indicate what the model is for.**

The critical epistemological question becomes whether the abstractions introduced by the model remain appropriate to the practice in which the model is employed.

5. The Problem of Verbalizing Thought

5.1. Think-Aloud and the Transparency Assumption

A second difficulty concerns methods used to investigate cognition. One influential approach is the analysis of verbal reports, including "Think Aloud" procedures in which subjects verbalize their thoughts while solving a problem.

Such methods can provide valuable information. However, verbalization introduces a methodological problem. The act of thinking and the act of describing one's thinking are not necessarily identical. When a subject is asked to verbalize cognition, the subject performs an additional operation: reflection upon the ongoing activity. The resulting report may therefore contain both elements of the original cognitive process and elements introduced by the act of reporting.

5.2. Description Is Not Identical with Action

Within the act-oriented framework, this distinction is not surprising. The subject who performs an act is situated within that act. The subject who subsequently describes the act occupies a partially different position. Description selects certain elements, organizes them linguistically, and presents them within a communicable structure.

Consequently, verbal reports may provide evidence about how subjects **reconstruct and conceptualize their own activity**, without providing transparent access to the activity in its entirety.

This does not make verbal reports useless. They remain valuable for studying: explicit reasoning, reflective strategies, metacognition, communicative practices, post hoc reconstruction, and consciously accessible procedures.

The limitation concerns the stronger claim that verbalization provides direct access to primary understanding. If understanding is fundamentally an act, then a verbal account is a representation of that act rather than the act itself.

6. An Alternative Methodology for the Study of Understanding

If conceptual understanding is an act, its investigation should not be restricted to descriptions of internal states. Other forms of evidence become important:

6.1. Successful Action — The first criterion is whether the subject can perform the relevant activity successfully. Understanding is manifested in competent orientation within the situation.

6.2. Error — Errors are especially informative because they reveal where the organization of experience breaks down. A subject may select the wrong feature, identify it incorrectly, or project an inappropriate consequence. The structure of the error can therefore provide information about the structure of the preceding act.

6.3. Breakdown — A previously successful act may cease to function when circumstances change. Such breakdowns expose the conditions under which the act was valid.

6.4. Transition — The movement from non-understanding to understanding provides another important object of investigation. Instead of asking only what representation the subject possesses after learning, one can examine how the organization of experience changes during learning.

This suggests an empirical program centered upon **transitions, errors, corrections, and successful adaptation**.

7. The Problem of Objectivity

7.1. The Subjectivity Objection

The act-oriented approach immediately encounters a classical objection. If cognition is understood as an activity performed by a subject in a particular context, does this not reduce knowledge to subjective perspective? If concepts depend upon situated acts, what prevents radically different subjects from constructing incompatible realities?

This objection is serious. An epistemological theory cannot simply replace correspondence with subjective usefulness and continue to claim objectivity without further argument. The act-oriented approach therefore requires a criterion that distinguishes arbitrary subjective success from forms of understanding capable of supporting objective knowledge.

8. Objectivity as Invariance Under Expansion of Practice

The proposed criterion is **invariance of the act under the expansion of practice**.

An act may initially be successful within a restricted context. Its epistemic status changes when that context is expanded. Suppose a conceptual orientation works for one observer, one situation, or one set of experimental conditions. That is evidence of local success.

Now introduce: new observers, new measurements, new environmental conditions, different goals, independent procedures, and additional cases. If the act continues to organize experience successfully, its claim to objectivity becomes stronger.

Objectivity is therefore approached not as absolute independence from the subject, but as **stability across an expanding field of practice**. This expansion includes the introduction of other subjects and their practices, not merely additional measurements. The broader the range of conditions in which the act remains successful, the stronger its claim to objectivity.

This proposal provides a non-metaphysical criterion for comparing conceptual orientations.

9. Objectivity Without Relativism

The proposed criterion must not be confused with relativism.

Relativism would imply that a conceptual framework is valid merely because it works for a particular individual or community. The present proposal requires the opposite. An act is epistemically stronger when its validity survives the introduction of conditions that could potentially undermine it.

Objectivity therefore contains an element of **resistance**. The world, other subjects, measurements, and practical consequences can resist a conceptual orientation. An interpretation that fails under expanded practice loses its claim to objectivity. An interpretation that continues to succeed acquires stronger epistemic standing.

This allows objectivity to remain connected with experience without making it identical to immediate usefulness.

10. The Example of Scientific Concepts

Consider the historical concept of phlogiston.

Within an earlier explanatory framework, it provided an intelligible account of combustion. However, as experimental practice expanded, including more precise weighing and analysis of gases, the explanatory structure became increasingly inadequate. Its problem was therefore not simply that another representation happened to replace it. Its conceptual organization ceased to remain successful under an expanded practice.

By contrast, concepts such as water retain their explanatory and practical relevance across radically different contexts.

The comparison illustrates the proposed criterion. A concept can be regarded as more objective when the act through which it organizes experience remains effective across a broader range of circumstances.

The example is not intended to establish the criterion conclusively. It illustrates the type of comparative analysis that the proposed framework makes possible.

11. Knowledge and Action

One of the principal consequences of the act-oriented framework concerns the relationship between knowledge and action.

Classical epistemology often permits knowledge and action to be represented as separate stages: *first one knows, then one acts*.

The act-oriented conception challenges this temporal and conceptual separation. To understand something is already to possess some capacity to orient oneself toward it. This does not mean that every successful action constitutes understanding. Rather, understanding is manifested in the ability to distinguish relevant features, identify them appropriately, anticipate consequences, and adjust action accordingly.

Knowledge is therefore not exhausted by information. Information can be possessed without understanding. Understanding becomes visible when information has been integrated into an effective organization of experience and action.

12. Tacit Understanding and Expertise

This framework also offers an interpretation of tacit knowledge.

Experts frequently perform actions that they cannot fully describe in propositional form. A skilled practitioner may recognize a pattern immediately, respond to a subtle change, or anticipate failure without being able to enumerate every element involved in the judgment.

This phenomenon is difficult to explain if knowledge is identified exclusively with explicit representations. It becomes more intelligible if conceptual understanding is treated as an act. The expert has acquired an organization of selection, identification, and projection through repeated engagement with situations. Expertise is therefore not simply a larger quantity of stored information. It is a transformed capacity for acting within a domain.

13. Education as Training of Conceptual Acts

The same argument has consequences for education.

If knowledge is primarily represented as the transmission of information, education is naturally conceived as the transfer of representations from teacher to learner. An act-oriented epistemology suggests a different model.

Education should create conditions in which learners repeatedly perform acts of selection, identification, projection, action, feedback, and correction. The learner does not merely receive the correct conceptual structure. The learner acquires the capacity to reconstruct such structures in new situations.

This makes project-based learning, problem-solving, simulation, laboratory practice, and other forms of situated activity philosophically significant. Their value is not merely motivational. They reproduce important structural features of understanding itself.

14. Relation to Existing Philosophical Traditions

The act-oriented paradigm should not be understood as an isolated alternative to the history of epistemology. It can instead be interpreted as an attempt to integrate several established insights:

- **Aristotelian philosophy** describes categories, distinctions, and definitions that can be understood as stabilized outcomes of conceptual activity.
- **Kantian philosophy** identifies structural conditions that make organized experience possible and thereby emphasizes the active contribution of cognition.
- **Pragmatism** connects meaning with practical consequences and therefore moves beyond purely contemplative conceptions of knowledge.
- **Phenomenology** investigates the structure of lived experience and thereby brings attention to cognition as situated rather than detached observation.
- **Enactivism** provides particularly close parallels because it understands cognition in terms of embodied interaction rather than isolated internal representation.
- **Cognitive science** provides empirical descriptions of mechanisms, representations, learning processes, and neural correlates associated with conceptual activity.

The present framework does not claim that these traditions are identical. Rather, it proposes that they may be interpreted as examining different dimensions of a broader phenomenon: **the organization of activity through which something becomes intelligible and actionable.**

15. The Epistemological Status of Representations

The transition from representation to act should therefore not be understood as a rejection of representation.

Representations remain indispensable. Scientific theories require symbolic systems. Language permits communication and reflection. Models allow complex relations to be stabilized. Diagrams make structures manipulable. Mathematical formalization allows certain operations to be reproduced and extended.

The act-oriented claim is more precise: **representation is not the whole of understanding.**

A representation becomes epistemically significant when incorporated into practices of recognition, prediction, testing, intervention, and correction. The same representation may therefore have different epistemic significance depending upon how it functions within practice. Its meaning cannot be determined solely by its internal structure.

16. Toward a Revised Epistemological Paradigm

The preceding analysis suggests several transformations:

- **From representation to orientation:** Knowledge is not only the possession of a representation but the capacity to orient activity.
- **From static structure to dynamic process:** Conceptuality is not exhausted by the structure that remains after an act has been stabilized.
- **From correspondence to operational constraint:** The relation between knowledge and reality is investigated through the resistance of experience to unsuccessful conceptual orientations.
- **From isolated cognition to situated activity:** Cognition is understood within concrete relations between subject, environment, history, and possible action.
- **From explicit knowledge to integrated competence:** Understanding includes forms of knowledge that may not be completely expressible as propositions.
- **From fixed objectivity to tested invariance:** Objectivity becomes a matter of stability under the expansion of practice.

These transformations do not abolish classical epistemology. They propose a different organization of its central problems.

17. A Preliminary Formal Schema

The act-oriented framework can be provisionally represented as:

Experience → Selection → Identification → Projection → Action → Feedback → Reorganization

The arrow from feedback to reorganization is essential. Without it, the framework would merely describe a linear sequence. With it, conceptual activity becomes potentially recursive. A successful act can stabilize a conceptual orientation. An unsuccessful act can force its modification. Repeated success across expanding conditions can increase its objective status. Repeated failure can undermine it.

This produces a dynamic epistemological structure: **conceptual understanding is not a fixed possession but a capacity capable of stabilization, application, testing, correction, and expansion.**

The schema should not be interpreted as a psychological algorithm. It is a theoretical representation of the structural relations proposed by the act-oriented conception.

18. Limits and Open Problems

The present framework remains preliminary. Several questions require further investigation:

1. The distinction between conceptual and non-conceptual acts must be clarified. If every successful orientation toward the environment is called a concept, the notion becomes excessively broad.
2. The relationship between conceptual acts and linguistic representations requires more precise analysis.
3. The proposed criterion of objectivity must be developed sufficiently to determine how different degrees of invariance can be compared.
4. The role of social practices requires further examination. Conceptual acts are not performed by isolated subjects alone; language, institutions, technologies, and communities contribute to the conditions under which acts become possible and intelligible.
5. Empirical methods are required to determine whether the proposed framework can generate testable predictions or methodological procedures.

These limitations should not be interpreted as defects that invalidate the proposal. They identify the research program that follows from it.

19. Conclusion: From Models to Acts of Understanding

The representational paradigm has provided indispensable resources for philosophy and science. Representations, models, formal systems, and neural descriptions remain necessary components of contemporary knowledge. Yet they do not by themselves explain the act through which something becomes meaningful in a concrete situation.

The act-oriented approach proposed here begins from a different premise. The concept is understood as an act of structuring experience through selection, identification, and projection.

From this premise, several epistemological consequences follow. Scientific models can be understood as stabilized instruments of orientation rather than literal copies of reality. The idealization inherent in models can therefore be interpreted as both a limitation and a condition of their practical usability. Verbal reports can be recognized as valuable but partial reconstructions of cognitive activity rather than transparent representations of primary understanding. Objectivity can be approached through the invariance of successful conceptual acts under the expansion of practice. Knowledge and action consequently appear not as completely separate domains but as mutually constitutive dimensions of understanding.

The central transition proposed by this paper can therefore be expressed as follows:

from representation to act, from correspondence to tested orientation, from information to competence, and from static objectivity to invariance across expanding practice.

This transition does not require abandoning science. On the contrary, it may provide a way of understanding why scientific representations, models, experiments, measurements, and theories are effective: they participate in practices through which conceptual acts are tested, stabilized, corrected, and extended.

The deepest implication of the act-oriented paradigm is therefore not that reality is constructed by the subject. It is that **knowledge becomes objective through the capacity of an act to survive encounters with an increasingly diverse reality of practice.**

Understanding is consequently neither the passive reception of information nor the possession of an internal picture. It is the acquired capacity to orient oneself successfully within a world that can confirm, resist, and transform one's conceptual activity.

The concept, in this sense, is not merely something through which we represent the world. It is an organized capacity through which **the world becomes actionable and intelligible.**

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